

UNIT-5 | Lecture 3

Query Equivalence

Coverage: Hour 3

Learning Objectives

Understand query equivalence.

Learn common equivalence rules used in query optimization.

Understand why equivalent expressions are useful.

Query Equivalence

Two relational algebra expressions are **equivalent** if they produce the same result for every valid database instance.

Equivalent expressions can have very different execution costs.

Common Equivalence Rules

Rule	Idea
Selection cascade	$\sigma_{c1} \text{ AND } c2(R)$ can be represented as $\sigma_{c1}(\sigma_{c2}(R))$
Selection commutativity	$\sigma_{c1}(\sigma_{c2}(R)) \equiv \sigma_{c2}(\sigma_{c1}(R))$
Projection cascade	Some projection sequences can be combined when required attributes are preserved
Join commutativity	$R \bowtie S \equiv S \bowtie R$ for an appropriate inner join
Join associativity	$(R \bowtie S) \bowtie T \equiv R \bowtie (S \bowtie T)$ under the corresponding join semantics

Selection Pushdown

If a selection condition uses attributes from only one relation, it can often be performed before a join.

Example idea: instead of joining all Student rows and then filtering Branch='CSE', first select CSE students and then join.

Why Equivalence Matters

It allows the optimizer to transform a query into a cheaper but logically equivalent expression.

The DBMS can choose a better execution plan without changing the query result.

Key Takeaway

Query equivalence provides the foundation for transforming expensive query expressions into cheaper equivalent forms.